ZCAS University

BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

CSS3211 SOFTWARE REQUIREMENTS ENGINEERING

FINAL EXAM

14:00 -17:00

Friday, 6th June 2025

TIME ALLOWED: THREE HOURS

INSTRUCTIONS:

1. Section A: ALL questions in this section are COMPULSORY and must be attempted
2. Section B: Answer any **THREE** questions only
3. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: All questions in this section are compulsory and must be answered

QUESTION 1

The University of TechZam is planning to develop a **University Student Management System (USMS)** to manage student records, course enrollments, and academic performance. The goal is to provide both administrative staff and students with a user-friendly system that can be accessed via web and mobile platforms.

The system should:

- Allow administrators to add, update, and delete student profiles.
- Enable students to register for courses and view academic results.
- Ensure data integrity, proper access control, and scalability.
- Use an object-oriented approach in both design and implementation.
- Include features like automated GPA calculations and report generation.

The development team is currently working on **gathering requirements, designing the system architecture**, and **implementing core modules** using object-oriented programming principles.

(a) List and describe **three major functional requirements** for the USMS. (6marks)

(b) Propose a **high-level architectural design** for the system. Briefly describe the components and how they interact. (6marks)

c) Identify **two non-functional requirements** that are critical for this system and explain why. (3 marks)

d) Identify **three key objects (classes)** in the USMS and list **two attributes and two methods** for each. (9 marks)

e) Explain the difference between **inheritance** and **encapsulation**, and describe how each could be applied in the system (6 marks)

f) Write a simple class definition in Python or Java for a Student object that includes the following:

Student Class();
int
var;

student_id, name, courses as attributes

Methods: **enroll_course(), view_results()** (6 marks)

g) Briefly explain how this class fits into the broader system design of USMS. (4 marks)

Students Class;
Students Class();

Students Class();
IOS string()

Students Class();
Students_Id

(40 marks)

SECTION B: Attempt any THREE questions in this section.

Question 2

"SafeRide" is a new ride-sharing app with the following key requirements:

1. **User Authentication:** Riders must log in using email/phone and a 6-digit PIN.
2. **Fare Calculation:** The fare must adjust in real-time based on distance, traffic, and surge pricing (max 2x base rate).
3. **Safety Feature:** Riders can trigger an SOS alert, which shares their live location with emergency contacts.
4. **Performance:** The app must respond to SOS requests within 5 seconds, even under low network conditions (2G).

a) Define **requirements validation** and explain how it differs from verification in this scenario. Why is traceability important when testing the Safety Feature requirement? (4 marks)

b) Give **one example** of a missing requirement that testing might reveal in the fare calculation logic. (2 marks)

c) **Design test cases** for the User Authentication requirement (including inputs, expected outputs). **Identify ambiguities** in the Fare Calculation requirement and rewrite it for clarity. (6 marks)

d) Propose a **testing technique** (e.g., boundary value analysis) to validate the Performance requirement for SOS alerts. (4 marks)

e) A tester reports that the SOS feature fails under 2G networks. Explain how you would. **Trace the defect** back to the original requirement.
Prioritize this bug versus a UI typo in the fare display (4 marks)

(20 marks)

Question 3 ✓

Design is the creative process of transforming the problem into a solution. The description of a solution is also known as design ie; the requirements specification defines the problem. The design document specifies a particular solution to the problem.

a) With the aid of a diagram illustrates the Conceptual and Technical Design. (10 marks)

b) Explain the conceptual design in view of the customer. (5 marks)

c) Explain the technical design in view of the programmer. (5 marks)

(20 marks)

Question 4

"HealthTrack" is a hospital management system being delivered to a regional clinic. Key requirements include:

1. **Patient Registration:** Capture demographics, medical history, and insurance details (mandatory fields).
2. **Appointment Scheduling:** Doctors can view/edit their schedules; patients receive SMS reminders.
3. **Data Security:** HIPAA compliance all patient data must be encrypted at rest and in transit.
4. **Uptime:** System must be available 99.9% during clinic hours (8 AM–8 PM).

The development team is preparing for final delivery.

a) **Define** user acceptance testing *(UAT)* and explain its role for HealthTrack. Why is a **Requirements Traceability Matrix (RTM)** critical before delivery? Provide a clinic-specific example. Identify **one risk** during data migration from the old system and how to mitigate it. (6 marks)

b) **The clinic insists on a phased rollout.** Design a 2-phase rollout plan for HealthTrack.
*** Justify your choice of modules for Phase 1**
(5 marks)

c) **Training** is required for clinical staff. Outline a training plan covering. Key user roles (doctors, receptionists).
How to handle system downtime. (5 marks)

d) A doctor reports that SMS reminders are delayed by 2 hours. Are these requirements **defect** or is it an **implementation bug**? Justify. How would you update the requirements document to prevent future issues? (4 marks)

(20 marks)

Question 5

You are part of a software engineering team tasked with developing a **custom Learning Management System (LMS)** for a mid-sized university. The LMS will support course registration, content delivery, student-teacher interaction, and online assessments. The university has emphasized the need for scalability and ease of use.

As the requirements engineer, you are expected to model the software process and define the software development life cycle (SDLC) that best fits the project needs.

a) Identify and describe **three key phases** of the software life cycle you would apply in this project. Explain the role each phase plays in addressing the university's needs.

(6 marks)

b) Choose a suitable **software process model** (e.g., Waterfall, Spiral, Agile, etc.) for this LMS project. Justify your choice by discussing at least **two advantages** and **one limitation** of the chosen model in the context of the university's requirements. (5 marks)

c) Using a **diagram or descriptive steps**, modeling the interaction between stakeholders (e.g., students, lecturers, administrators) and system requirements gathering during the early phase of the process. (5 marks)

d) Explain how requirements validation and management are integrated into the life cycle model you selected. Provide **one example** of how failing to validate a requirement early could impact the LMS project. (4 marks)

(20 marks)

END OF EXAM